Experiment 8

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Branch: CSE **Section/Group: KRG 3-B**
Semester: 6th **Date of Performance:31/03/2026**
Subject Name: Full Stack Development – II **Subject Code: 23CSH-309**

1. **Aim**: To design and implement a Spring Boot application that integrates third-party APIs including **Stripe Payment Gateway** for handling online transactions and **Gmail SMTP Service** for sending email notifications after successful payments.

2. **Objective**:
   - To understand the concept of third-party API integration
   - To integrate Stripe Payment Gateway for secure online payments
   - To configure and use Gmail SMTP for sending emails
   - To implement REST APIs for payment processing
   - To handle payment success and failure responses
   - To send automated email notifications after successful payment
   - To follow Layered Architecture (Controller – Service – Repository)
   - To enhance real-world application functionality using external services

3. **Implementation / Code:**

- **Tools & Technologies Used:-**

  - Java
  - Spring Boot
  - Spring Web
  - Spring Data JPA
  - Maven
  - IntelliJ IDEA
  - Stripe API
  - Gmail SMTP (Java Mail Sender)
  - Postman / Browser
  - PostgreSQL Database

- **Implementation Description:-**

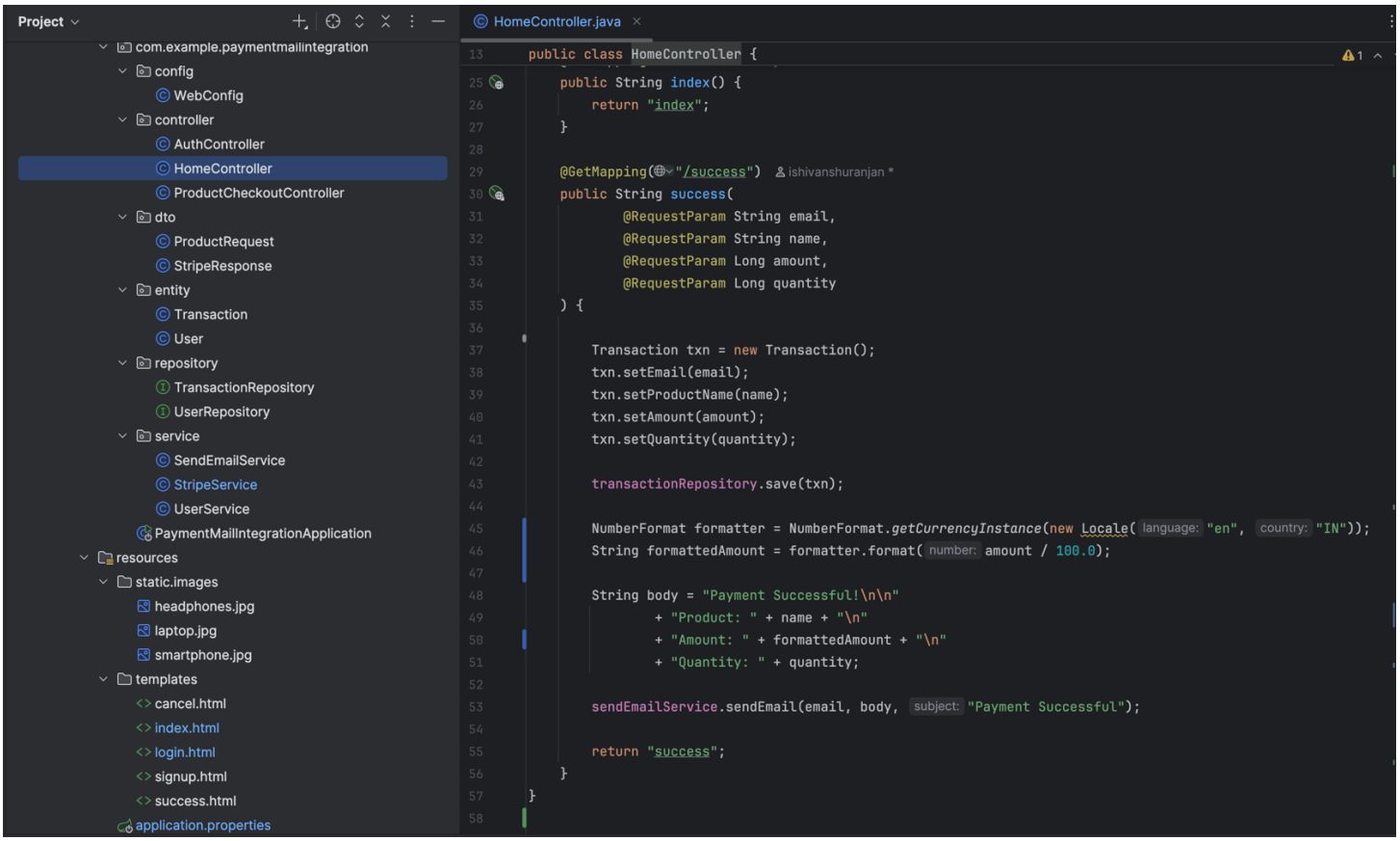
  - A Spring Boot project named Payment Mail Integration is created.

  - The project integrates Stripe API for handling online payments securely.

  - Users enter card details on Stripe Checkout page and complete payment.

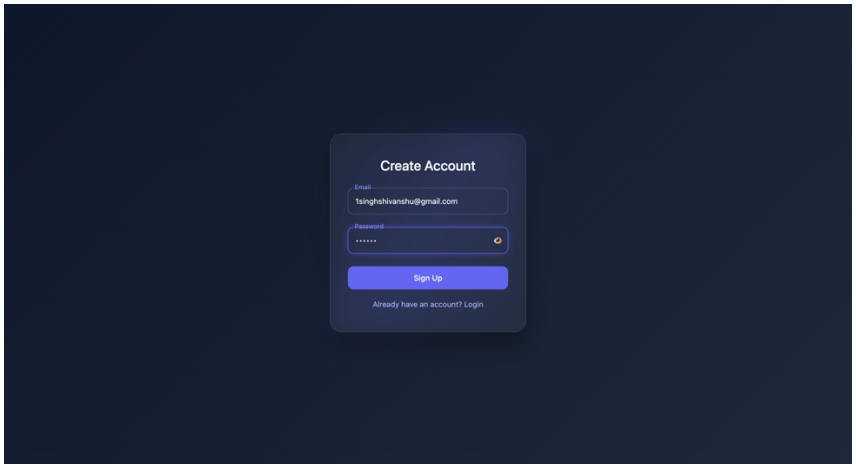
- After successful payment, a success page is displayed.
- A confirmation email is automatically sent to the user using Gmail SMTP.

- **Project Structure:-**



4. Output:

- Spring Boot application started successfully on **localhost:8080**
- Stripe payment page loaded successfully
- Payment completed successfully using test card
- Success page displayed after payment completion
- Email notification sent successfully to registered email
- Email contains product details, amount, and quantity
- Payment flow tested successfully using Stripe test environment
- Application integrated successfully with third-party APIs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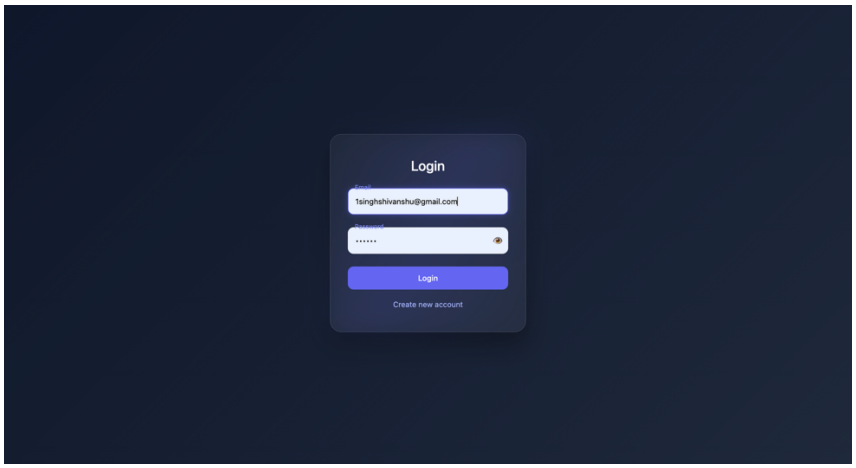
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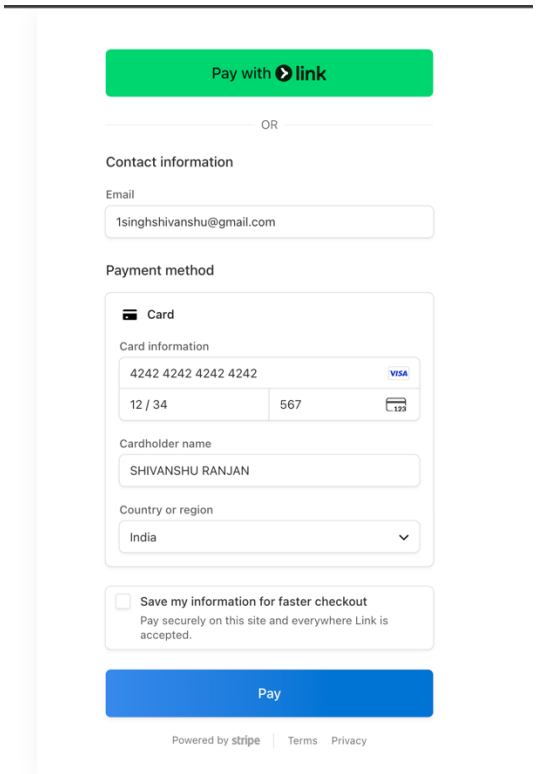
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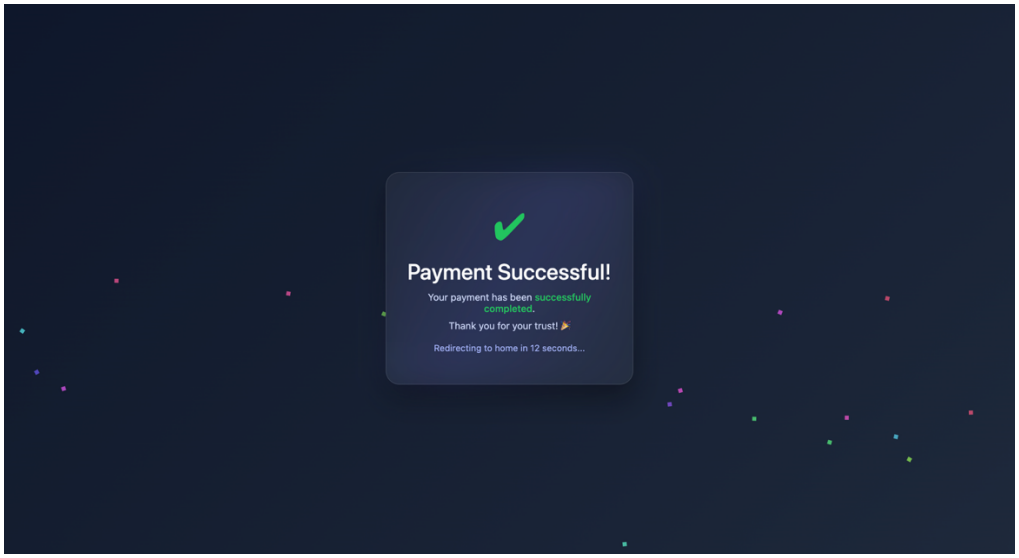

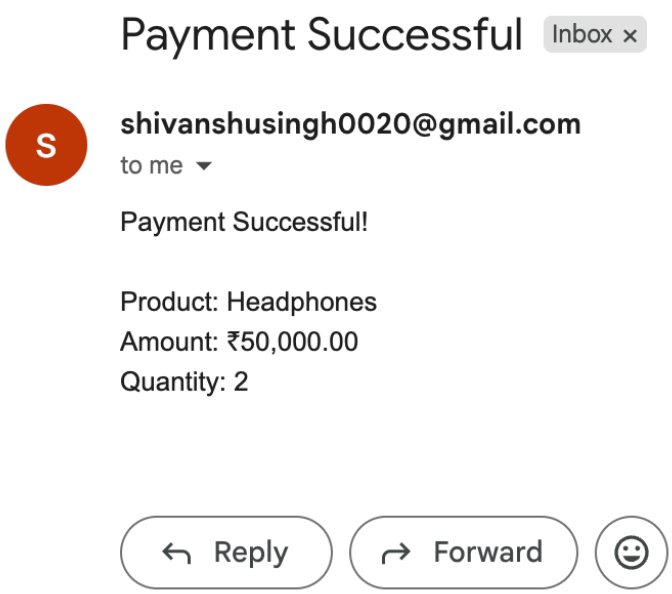

5. Learning Outcomes (What I Have Learnt):

- Understanding of third-party API integration
- Working with Stripe Payment Gateway
- Secure handling of online transactions
- Sending emails using Gmail SMTP
- Implementing real-world payment systems
- Integration of frontend and backend systems
- Usage of REST APIs in real applications
- Handling success and failure scenarios
- Improving application functionality using external services